

Lab 1 – Smart Lab Equipment & Slot Reservation Portal

Use-Case Flow Specification

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Use Case ID	UC-03
Use Case Name	Reserve Equipment Slot
Primary Actor	Student
Secondary Actor(s)	Lab Technician (indirectly, via equipment calibration records maintained beforehand)
Related Requirements	FR-001 (reservation rules), FR-002 (calibration check), NFR-001 (concurrency & lock isolation)
Includes	«include» UC-01 Authenticate User; «include» UC-05 Verify Equipment Calibration Status

Preconditions

- The Student is a registered, authenticated user of the portal (or successfully completes login via the included Authenticate User use case).
- **At least one unit of the requested equipment exists in the catalog.**
- The Student has no active late-return disciplinary restriction on their account (see FR-005).

Postconditions

- **Success:** A reservation record is created and locked for the selected equipment and time slot (≤ 2 hours, ≤ 7 days in advance); a unique reservation token is generated and shown to the Student.
- **Failure:** No reservation record is created; the equipment slot remains available for other students; the Student is shown a reason for rejection.

Main Success Scenario

1. The Student logs into the portal. [*«include» UC-01 Authenticate User*]
2. The Student navigates to the equipment catalog. The system displays real-time availability for all lab instruments (oscilloscopes, logic analyzers, FPGA development boards).
3. The Student selects an equipment item and specifies a desired date, start time, and duration (≤ 2 hours), within the next 7 days.
4. The system verifies that the selected equipment's calibration status is "Calibrated." [*«include» UC-05 Verify Equipment Calibration Status*]
5. The system checks the requested slot against existing reservations for conflicts, applying database-level lock isolation so concurrent requests are serialized safely.
6. The system locks the slot, creates the reservation record, and generates a unique reservation token within 200 ms of the request.
7. The system displays a confirmation screen to the Student showing the reservation token, equipment name, and slot timing.
8. The use case ends successfully.

Alternate Flows

A1 – Equipment Not Calibrated (extends Step 4)

- 4a1. The system finds the equipment's calibration status is "Due for Calibration" or "Under Maintenance."
- 4a2. The system rejects the reservation attempt and displays a message stating the equipment is unavailable for booking.
- 4a3. The system suggests alternative calibrated equipment, if any.
- 4a4. The use case ends without a reservation being created.

A2 – Slot Conflict / Double-Booking Attempt (extends Step 5)

- 5a1. The system detects that the requested slot overlaps with an existing locked reservation.
- 5a2. The system displays a "Slot Unavailable" message and refreshes the availability view with the current open slots.
- 5a3. The Student selects a different date/time, or ends the use case without reserving.

A3 – Student Under Disciplinary Restriction (extends Step 1): The system detects an active late-return disciplinary restriction on the Student's account (FR-005 / UC-09 Restrict Reservation Privileges), blocks access to the reservation flow, displays the reason for restriction, and the use case ends without a reservation being created.